

Lab10: Structural Bioinformatics

Candice Lei (A18585167)

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Introduction to the RCSB Protein Data Bank (PDB)

```
pdb <- read.csv("pdbstats26")
names(pdb)
```

```
[1] "Molecular.Type"  "X.ray"           "EM"              "NMR"
[5] "Integrative"     "Multiple.methods" "Neutron"         "Other"
[9] "Total"
```

```
head(pdb)
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178795	21825	12773	343	226
2	Protein/Oligosaccharide	10363	3564	34	8	11
3	Protein/NA	9106	6335	287	24	7
4	Nucleic acid (only)	3132	221	1566	3	15

5		Other	175	25	33	4	0
6	Oligosaccharide (only)		11	0	6	0	1
	Neutron	Other	Total				
1	84	32	214078				
2	1	0	13981				
3	0	0	15759				
4	3	1	4941				
5	0	0	237				
6	0	4	22				

```
total_structures <- sum(pdb$Total)
total_structures
```

```
[1] 249018
```

Q1. What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

X-ray: 80.95% and Electron Microscopy: 12.84%.

```
xray_percent <- sum(pdb$X.ray) / total_structures * 100
em_percent <- sum(pdb$EM) / total_structures * 100

round(xray_percent, 2)
```

```
[1] 80.95
```

```
round(em_percent, 2)
```

```
[1] 12.84
```

Q2. What proportion of structures in the PDB are protein?

The proportion is 0.979 or about 97.92%.

```
protein_rows <- grepl("Protein", pdb$Molecular.Type)
protein_proportion <- sum(pdb$Total[protein_rows]) / total_structures

round(protein_proportion, 3)
```

```
[1] 0.979
```

```
round(protein_proportion * 100, 2)
```

```
[1] 97.91
```

Q3. Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 4,998 HIV-1 protease structures in the current PDB.

Visualizing the HIV-1 Protease Structure

Q4. Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

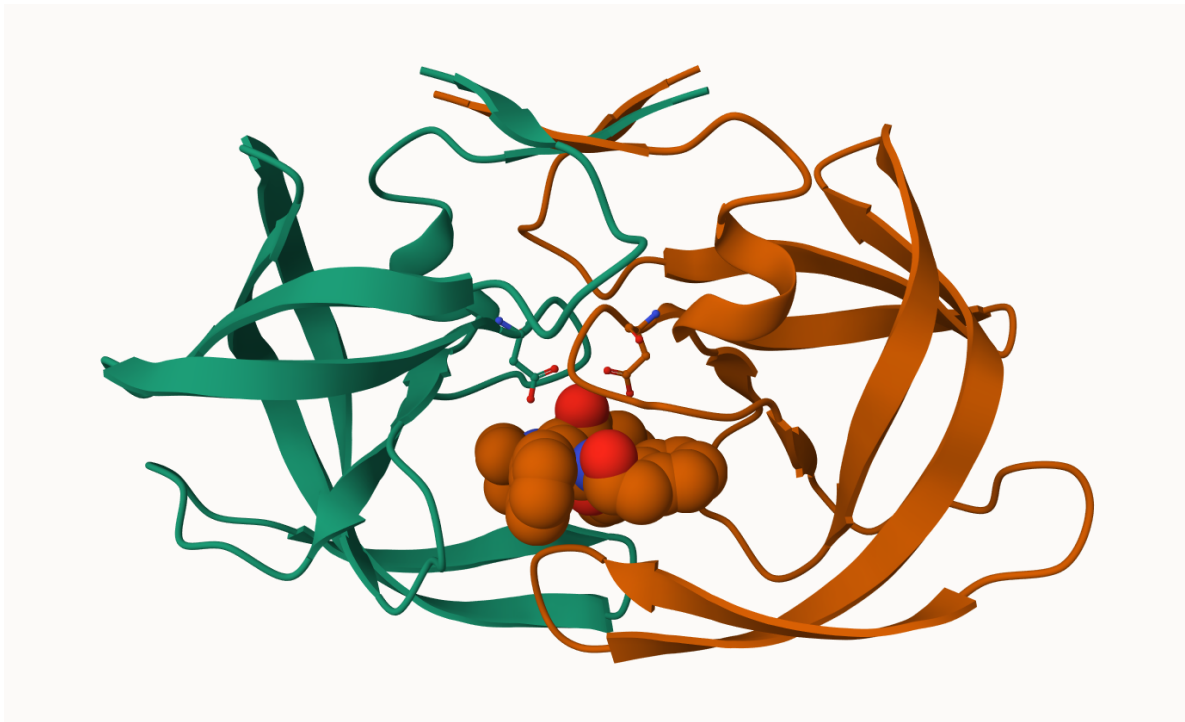
We see just one atom per water molecule in this structure because it only shows the oxygen atom of each water molecule. In X-ray crystallography, hydrogen are not usually visible.

Q5. There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

Yes, I can identify the water molecule. The residue number is HOH 308.

Q6. Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

```
knitr::include_graphics("1HSG-5.png")
```



Introduction to Bio3D in R

```
library(bio3d)
pdb <- read.pdb("1HSG.pdb")
pdb
```

Call: `read.pdb(file = "1HSG.pdb")`

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)

Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7. How many amino acid residues are there in this pdb object?

There are 198 amino acid residues.

Q8. Name one of the two non-protein residues?

MK1

Q9. How many protein chains are in this structure?

There are 2 protein chains in this structure.

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

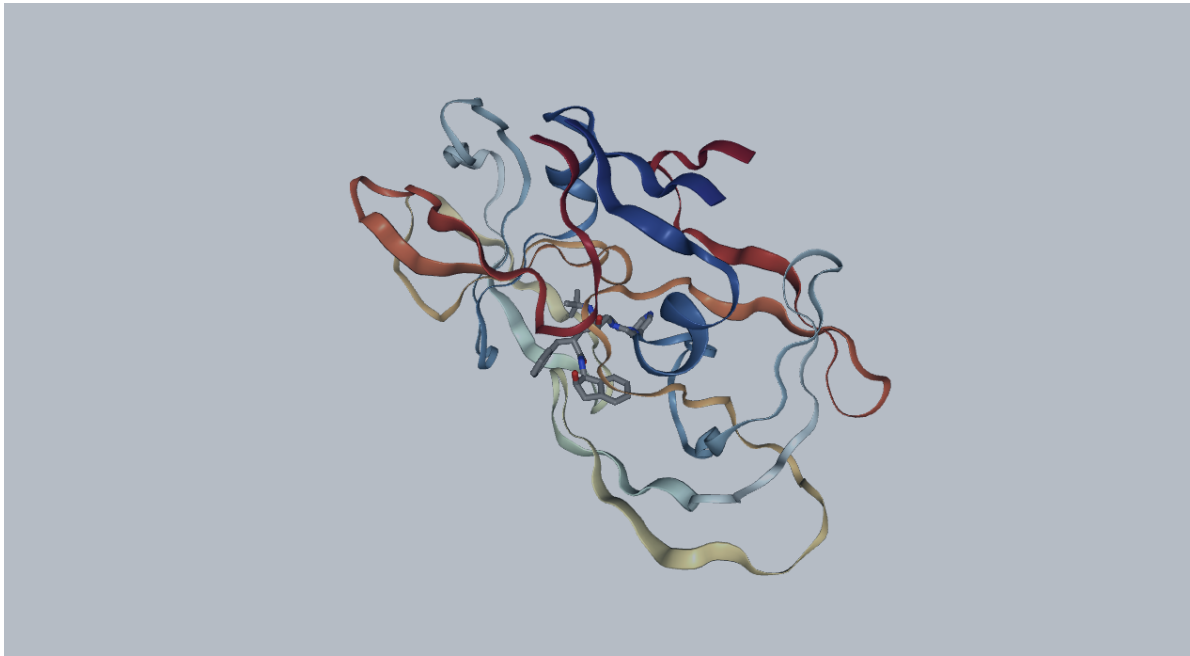
	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

```
# library(bio3dview)
# library(NGLViewerR)

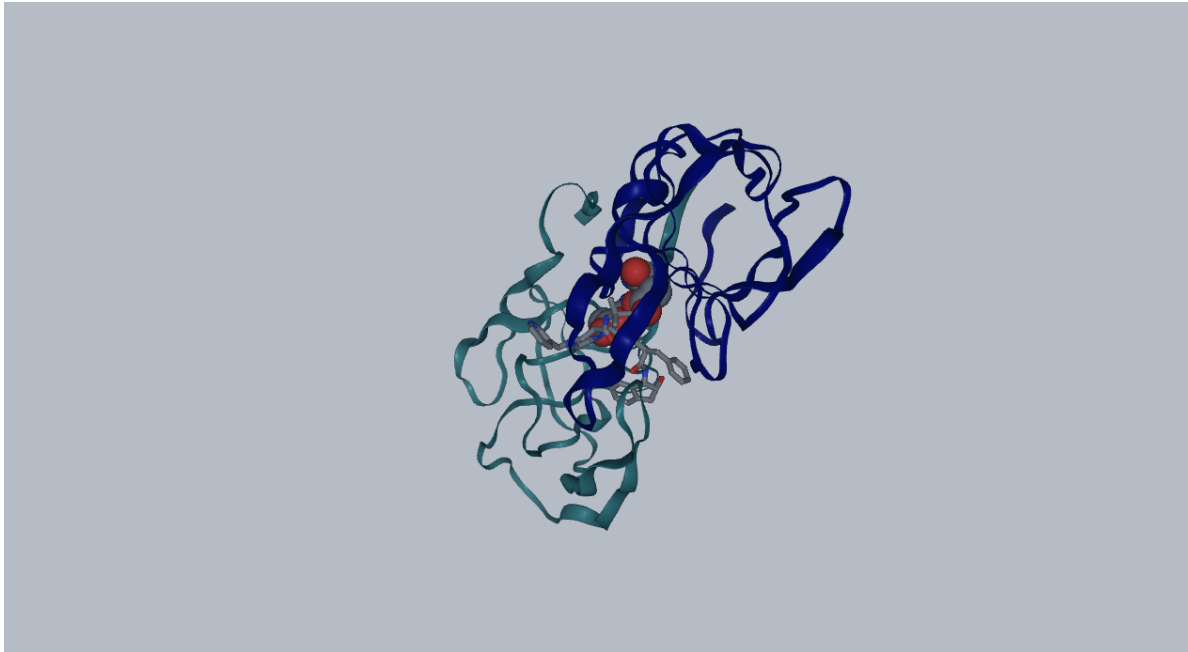
# view.pdb(pdb) |>
# setSpin()

knitr::include_graphics("setspin.png")
```



```
# sele <- atom.select(pdb, resno=25)
# view.pdb(pdb, cols=c("navy","teal"),
#         highlight = sele,
#         highlight.style = "spacefill") |>
# setRock()

knitr::include_graphics("rock.png")
```



```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
 PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT  

DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDKI
```

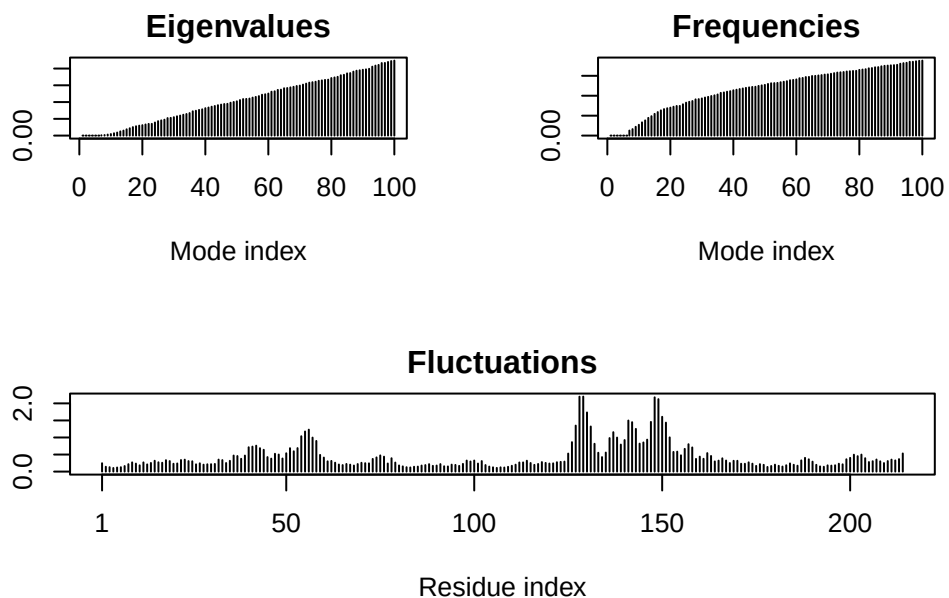
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
 YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

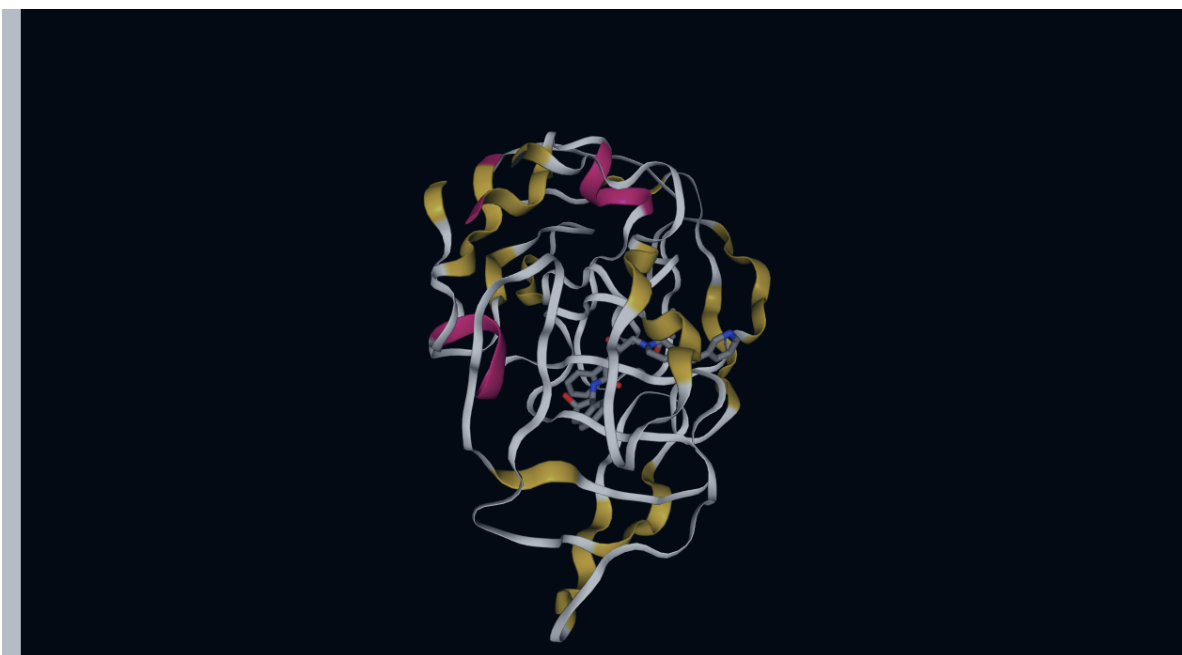
```
Building Hessian...      Done in 0.008 seconds.  
Diagonalizing Hessian... Done in 0.167 seconds.
```

```
plot(m)
```



Let's try a custom view

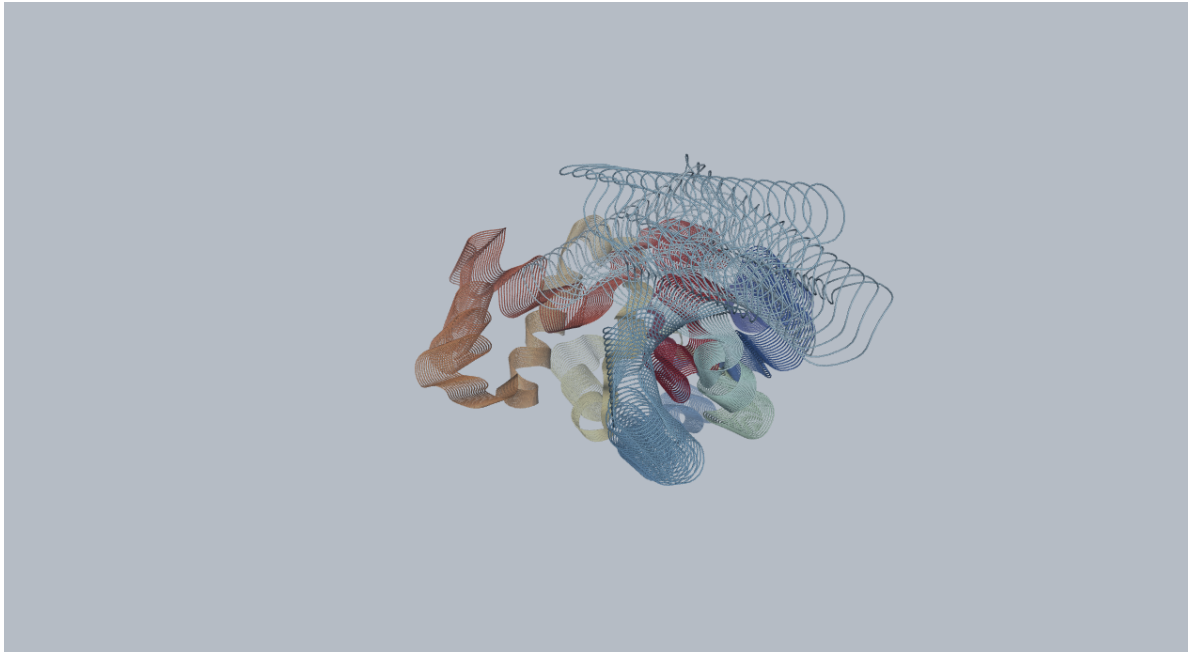
```
# view.pdb(pdb,  
#           colorScheme = "sse",  
#           backgroundColor = "black")  
  
knitr::include_graphics("view.pdb.png")
```

```
mktrj(m, file="adk_m7.pdb")
```

```
# view.nma(m, pdb=adk)
```

```
knitr::include_graphics("nma.png")
```



Comparative Structure Analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?

The msa package is only found in BioConductor.

Q11. Which of the above packages is not found on BioConductor or CRAN?

The bio3dview is not found.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True

Search and Retrieve ADK Structures

```
library(bio3d)
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      60
pdb|1AKE|A MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLVT
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A DELVIALVKERIAQEDCRNGFLLDGFRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120

     121      .      .      .      .      .      180
pdb|1AKE|A VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

+ attr: id, ali, call

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in this sequence.

```
hits <- NULL
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A', '6H
# Download related PDB files
files <- get.pdb(hits$ pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

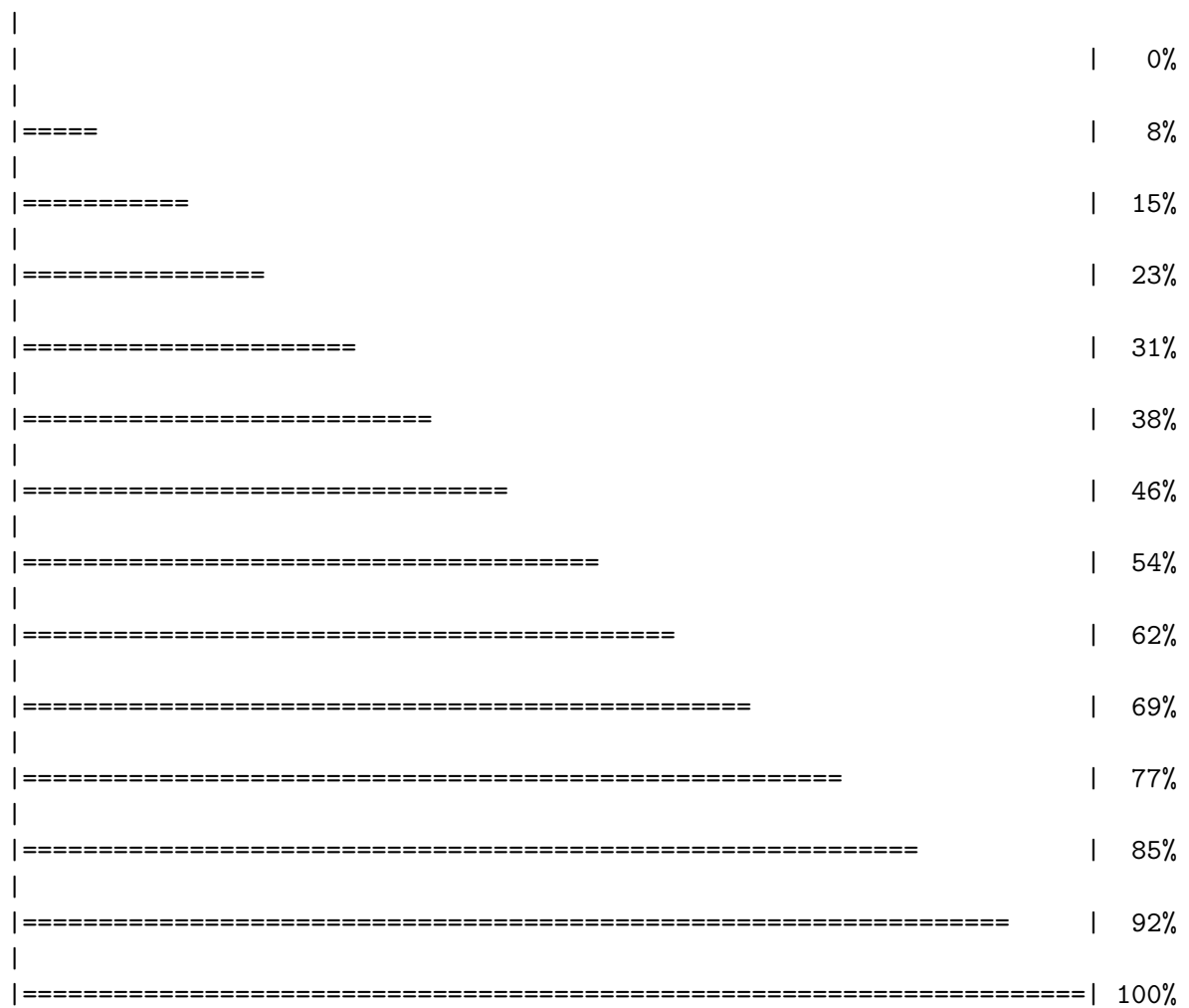
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download



```
# Align releated PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
```

```

pdbc/split_chain/6HAP_A.pdb
pdbc/split_chain/6HAM_A.pdb
pdbc/split_chain/4K46_A.pdb
pdbc/split_chain/3GMT_A.pdb
pdbc/split_chain/4PZL_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...

```

Extracting sequences

```

pdb/seq: 1   name: pdbc/split_chain/1AKE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbc/split_chain/6S36_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbc/split_chain/6RZE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbc/split_chain/3HPR_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbc/split_chain/5EJE_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 10  name: pdbc/split_chain/6HAM_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdbc/split_chain/4K46_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbc/split_chain/3GMT_A.pdb
pdb/seq: 13  name: pdbc/split_chain/4PZL_A.pdb

```

Annotate Collected PDB Structures

```

# Vector containing PDB database codes
ids <- basename.pdb(pdbc$id)

```

```
anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-
ray					
6S36_A	6S36	A	Protein	214	X-
ray					
6RZE_A	6RZE	A	Protein	214	X-
ray					
3HPR_A	3HPR	A	Protein	214	X-
ray					
1E4V_A	1E4V	A	Protein	214	X-
ray					
5EJE_A	5EJE	A	Protein	214	X-
ray					
1E4Y_A	1E4Y	A	Protein	214	X-
ray					
3X2S_A	3X2S	A	Protein	214	X-
ray					
6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					
4K46_A	4K46	A	Protein	214	X-
ray					
3GMT_A	3GMT	A	Protein	230	X-
ray					
4PZL_A	4PZL	A	Protein	242	X-
ray					
	resolution	scopDomain			pfam

1AKE_A	2.00	Adenylate kinase	Adenylate kinase (ADK)
6S36_A	1.60	<NA>	Adenylate kinase (ADK)
6RZE_A	1.69	<NA>	Adenylate kinase (ADK)
3HPR_A	2.00	<NA>	Adenylate kinase (ADK)
1E4V_A	1.85	Adenylate kinase	Adenylate kinase (ADK)
5EJE_A	1.90	<NA>	Adenylate kinase (ADK)
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase (ADK)
3X2S_A	2.80	<NA>	<NA>
6HAP_A	2.70	<NA>	Adenylate kinase (ADK)
6HAM_A	2.55	<NA>	Adenylate kinase (ADK)
4K46_A	2.01	<NA>	Adenylate kinase (ADK)
3GMT_A	2.10	<NA>	<NA>
4PZL_A	2.10	<NA>	Adenylate kinase, active site lid (ADK_lid)

ligandId

1AKE_A	AP5
6S36_A	CL (3),NA,MG (2)
6RZE_A	NA (3),CL (2)
3HPR_A	AP5
1E4V_A	AP5
5EJE_A	AP5,C0
1E4Y_A	AP5
3X2S_A	JPY (2),AP5,MG
6HAP_A	AP5
6HAM_A	AP5
4K46_A	ADP,AMP,P04
3GMT_A	S04 (2)
4PZL_A	CA,FMT,GOL

ligandName

1AKE_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6S36_A	CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A	SODIUM ION (3),CHLORIDE ION (2)
3HPR_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
1E4V_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
5EJE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
3X2S_A	N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6HAM_A	BIS(ADENOSINE)-5'-

PENTAPHOSPHATE

4K46_A ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A SULFATE ION (2)
4PZL_A CALCIUM ION,FORMIC ACID,GLYCEROL

source

1AKE_A Escherichia coli
6S36_A Escherichia coli
6RZE_A Escherichia coli
3HPR_A Escherichia coli K-12
1E4V_A Escherichia coli
5EJE_A Escherichia coli 0139:H28 str. E24377A
1E4Y_A Escherichia coli
3X2S_A Escherichia coli str. K-12 substr. MDS42
6HAP_A Escherichia coli 0139:H28 str. E24377A
6HAM_A Escherichia coli K-12
4K46_A Photobacterium profundum
3GMT_A Burkholderia pseudomallei 1710b
4PZL_A Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBIT

6S36_A

6RZE_A

3HPR_A

1E4V_A

loop

5EJE_A

1E4Y_A

loop

3X2S_A

conjugated adenylate kinase

6HAP_A

6HAM_A

4K46_A

3GMT_A

4PZL_A

Cryst

The crys

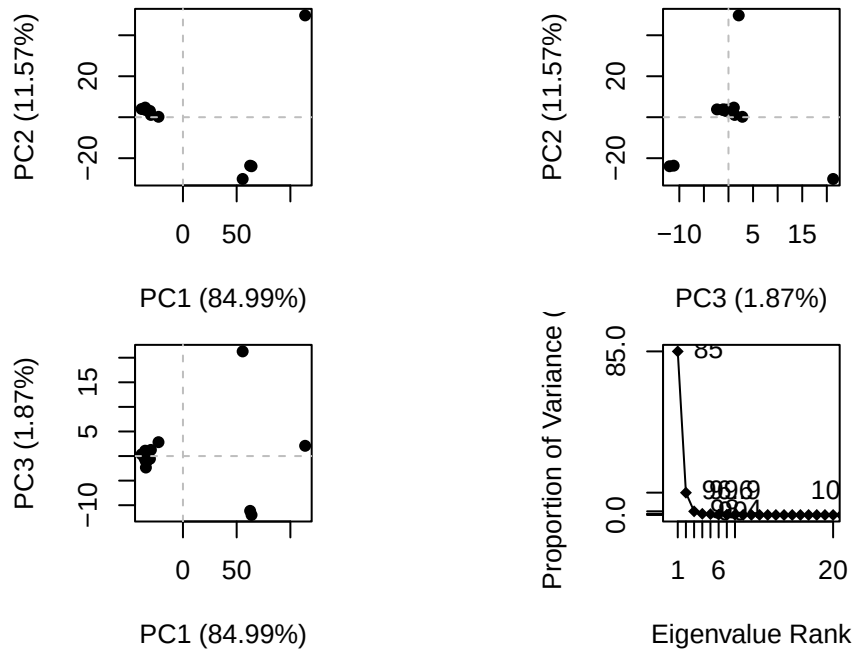
		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al.	J Mol Biology (1992)	0.19600	NA
6S36_A	Rogne, P., et al.	Biochemistry (2019)	0.16320	0.23560
6RZE_A	Rogne, P., et al.	Biochemistry (2019)	0.18650	0.23500
3HPR_A	Schrank, T.P., et al.	Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al.	Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al.	Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al.	Proteins (1993)	0.17800	NA

3X2S_A	Fujii, A., et al. Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al. Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al. To be published	0.19360	0.23680

	rWork	spaceGroup
1AKE_A	0.19600	P 21 2 21
6S36_A	0.15940	C 1 2 1
6RZE_A	0.18190	C 1 2 1
3HPR_A	0.20620	P 21 21 2
1E4V_A	0.19600	P 21 2 21
5EJE_A	0.18630	P 21 2 21
1E4Y_A	0.17800	P 1 21 1
3X2S_A	0.20700	P 21 21 21
6HAP_A	0.22370	I 2 2 2
6HAM_A	0.20311	P 43
4K46_A	0.16730	P 21 21 21
3GMT_A	0.23500	P 1 21 1
4PZL_A	0.19130	P 32

Principal Component Analysis

```
# Perform PCA
pc.xray <- pca(pdbx)
plot(pc.xray)
```

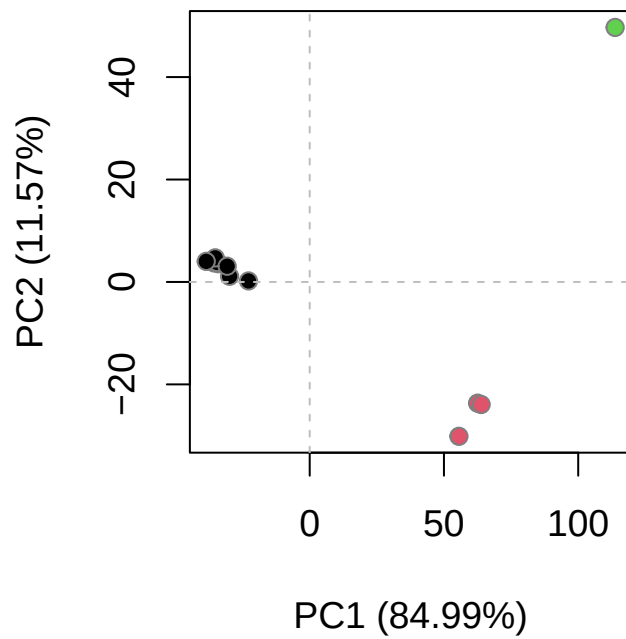


```
# Calculate RMSD
rd <- rmsd(pdb)
```

Warning in rmsd(pdb): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



PCA Visualization

```
# Visualize first principal component  
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2  
library(ggplot2)  
library(ggrepel)  
  
df <- data.frame(PC1=pc.xray$z[,1],  
                 PC2=pc.xray$z[,2],  
                 col=as.factor(grps.rd),  
                 ids=ids)  
  
p <- ggplot(df) +  
  aes(PC1, PC2, col=col, label=ids) +  
  geom_point(size=2) +  
  geom_text_repel(max.overlaps = 20) +  
  theme(legend.position = "none")  
p
```

